

# SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

## ASSESSMENT TOOL 2 – CONCEPT MAPPING

|                         |                                                                                                                                                                |                      |                                     |
|-------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------|-------------------------------------|
| <b>Course Code:</b>     | CSA06                                                                                                                                                          | <b>Course Title:</b> | Design and Analysis of Algorithms   |
| <b>CO Assessed:</b>     | CO1 – Precision (BL1, BL2, BL3)                                                                                                                                | <b>Assessment:</b>   | Assessment Tool 2 – Concept Mapping |
| <b>Weightage:</b>       | 20% of CIA                                                                                                                                                     | <b>Total Marks:</b>  | 100 (2 Questions × 50 Marks)        |
| <b>CO1 Statement:</b>   | <i>Determine algorithm efficiency using asymptotic notations and mathematical techniques including Master Theorem and substitution method. (BL1, BL2, BL3)</i> |                      |                                     |
| <b>Student Name:</b>    | R Saiteja                                                                                                                                                      | <b>Reg. No.:</b>     | 192424352                           |
| <b>Section / Batch:</b> | C                                                                                                                                                              | <b>Date:</b>         | 22.07.2026                          |

### Instructions to Students

- Answer BOTH questions. Each question carries 50 Marks. Total: 100 Marks.
- Draw a clear and neat concept map for each question.
- Identify all key concepts and arrange them in a logical hierarchy.
- Show meaningful linkages between concepts using arrows and connecting words.
- Compare related concepts wherever required and justify the relationships clearly.
- Ensure the concept map is well-organized, readable, and properly labelled.

|                        |                                                                                                                                 |         |                  |
|------------------------|---------------------------------------------------------------------------------------------------------------------------------|---------|------------------|
|                        |                                                                                                                                 |         |                  |
| <b>Concept Mapping</b> | Map algorithm efficiency concepts using asymptotic analysis, search techniques, recurrence relations, and recursion tree method | Q1 – Q2 | Q1 –50<br>Q2 –50 |
|                        |                                                                                                                                 |         |                  |

### SECTION A – CONCEPT MAPPING

#### Code of Conduct

*I B. Chandu (192525289) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the Student: B. Chandu

### RUBRICS FOR CALCULATION

|  |  |  |  |  |  |
|--|--|--|--|--|--|
|  |  |  |  |  |  |
|--|--|--|--|--|--|

|                        |     |                                                                            |                                                   |                                                             |                                         |
|------------------------|-----|----------------------------------------------------------------------------|---------------------------------------------------|-------------------------------------------------------------|-----------------------------------------|
| Concept Identification | 25% | Identifies all key concepts from literature accurately and comprehensively | Identifies most key concepts with minor omissions | Identifies limited concepts; some are irrelevant or missing | Fails to identify essential concepts    |
| Linkages               | 25% | Establishes clear, meaningful, and accurate relationships between concepts | Shows correct relationships with minor gaps       | Relationships are weak, unclear, or partially incorrect     | No meaningful relationships established |

|                               |                                                                                                                                                         |                                                                                                           |
|-------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------|
| Organization 20%              | <p>Concepts are neat, visually structured, and easy to interpret</p> <p><b>Evaluation:</b> Mostly organized with minor structural issues</p>            | <p>minor readability issues</p> <p>No logical organization or structure</p>                               |
| Integration of Literature 20% | <p>well-structured hierarchically with logical flow and grouping</p> <p>Integrates multiple studies effectively to show connections and comparisons</p> | <p>Limited integration; mostly isolated concepts</p> <p>No integration of literature evidence</p>         |
| Clarity 10%                   | <p>Shows some integration of literature</p> <p>Concept map is clear, Generally clear with</p>                                                           | <p>Clarity is reduced; difficult to interpret in parts</p> <p>Poorly presented and hard to understand</p> |

| Question No.        | Concept Identification (25%) | Linkages (25%) | Organization (20%) | Integration of Literature (20%) | Clarity (10%) | Total Marks (Each – 50) |
|---------------------|------------------------------|----------------|--------------------|---------------------------------|---------------|-------------------------|
| Q1                  |                              |                |                    |                                 |               |                         |
| Q2                  |                              |                |                    |                                 |               |                         |
| Overall Total (100) |                              |                |                    |                                 |               |                         |

| Faculty In-charge               | Course Coordinator              | Head of Department              |
|---------------------------------|---------------------------------|---------------------------------|
|                                 |                                 |                                 |
| Signature: _____<br>Date: _____ | Signature: _____<br>Date: _____ | Signature: _____<br>Date: _____ |

| S.No | Register No. | Student Name                           | Assigned Question 1 | Assigned Question 2 |
|------|--------------|----------------------------------------|---------------------|---------------------|
| 1    | 192311215    | MUTHU VAISHNAVI T                      | Q1                  | Q2                  |
| 2    | 192311229    | JENIECATHRINA J                        | Q3                  | Q4                  |
| 3    | 192324152    | ABINAYA SRI N                          | Q5                  | Q6                  |
| 4    | 192324287    | PRAVEEN KUMAR R                        | Q7                  | Q8                  |
| 5    | 192372122    | NADENDLA THANVEER BASHA                | Q9                  | Q10                 |
| 6    | 192372336    | SIGI SREENIVASULU                      | Q11                 | Q12                 |
| 7    | 192411025    | SRINIVASAN A                           | Q13                 | Q14                 |
| 8    | 192411031    | GUDIPALLI VEMULA SHABARISH             | Q15                 | Q16                 |
| 9    | 192411039    | GOKUL RAJAN R                          | Q17                 | Q18                 |
| 10   | 192411061    | GANGADEVI LOKESH                       | Q19                 | Q20                 |
| 11   | 192411156    | KEERTHIPATI HEMA SREE                  | Q21                 | Q22                 |
| 12   | 192411249    | CHENNABOINA BHAVYA SRI                 | Q23                 | Q24                 |
| 13   | 192421030    | NAVEEN KUMAR S                         | Q25                 | Q26                 |
| 14   | 192421126    | MOHAMMED ANAS M                        | Q27                 | Q28                 |
| 15   | 192421259    | JAYAKRISHNA V                          | Q29                 | Q30                 |
| 16   | 192421333    | PUSHPAVALLI V                          | Q31                 | Q32                 |
| 17   | 192424016    | DINESH M S                             | Q33                 | Q34                 |
| 18   | 192424174    | YOKESH KANNA R                         | Q35                 | Q36                 |
| 19   | 192424259    | PERA HEMIKA                            | Q37                 | Q38                 |
| 20   | 192424327    | CHARULATHA S                           | Q39                 | Q40                 |
| 21   | 192424352    | R SAI TEJA                             | Q41                 | Q42                 |
| 22   | 192465048    | POTHURAJU SAI CHARAN TEJ               | Q43                 | Q44                 |
| 23   | 192472087    | MANCHIKALAPATI VISHNU VARDHAN          | Q45                 | Q46                 |
| 24   | 192472149    | CHINNI VENKATA NAGA SAI<br>POORNA TEJA | Q47                 | Q48                 |
| 25   | 192472307    | SHAIK AZEEM                            | Q49                 | Q50                 |
| 26   | 192511060    | Manoj M                                | Q51                 | Q52                 |
| 27   | 192521075    | Kaviya R                               | Q53                 | Q54                 |
| 28   | 192521121    | Srimathi S                             | Q55                 | Q56                 |
| 29   | 192521352    | Jagadish V                             | Q57                 | Q58                 |
| 30   | 192524078    | Rafeez Ahamed S I                      | Q59                 | Q60                 |
| 31   | 192524237    | Athin Siva S                           | Q61                 | Q62                 |

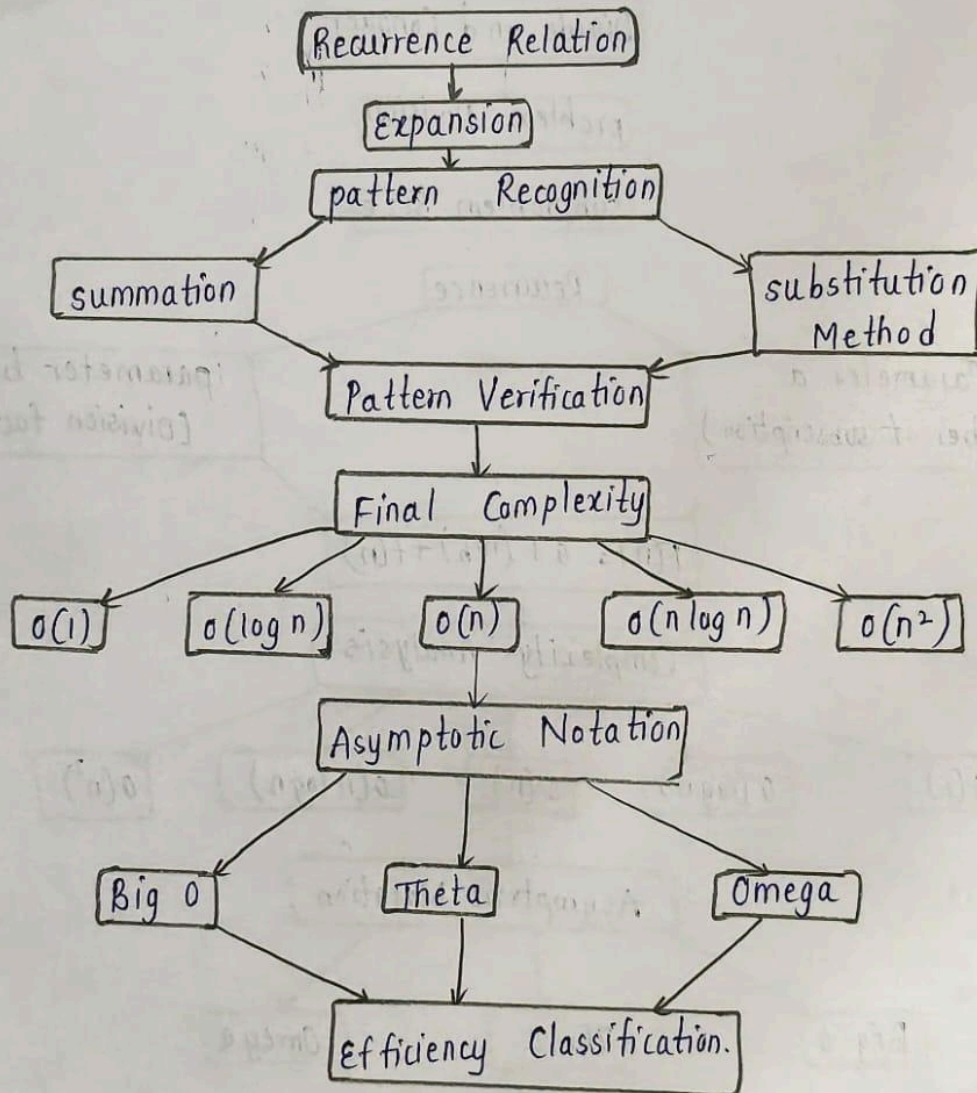
|    |           |                          |     |     |
|----|-----------|--------------------------|-----|-----|
| 32 | 192525202 | Muthu Neelakanta Naidu   | Q63 | Q64 |
| 33 | 192525289 | B Chandu                 | Q65 | Q66 |
| 34 | 192525303 | Yedururu Jithendra Reddy | Q67 | Q68 |
| 35 | 192565016 | Divya Sree P             | Q69 | Q70 |
| 36 | 192565069 | Yaswanthika Shree S      | Q71 | Q72 |
| 37 | 192571005 | V Sanjana                | Q73 | Q74 |
| 38 | 192571047 | Shaffria A               | Q75 | Q76 |
| 39 | 192571051 | Podalapalli Darshith     | Q77 | Q78 |
| 40 | 192572106 | Theshon Mishmaan M       | Q79 | Q80 |

1). Consider the concept map: Recurrence Relation → Expansion → Pattern Recognition → Summation → Final Complexity Explain how recurrence patterns are identified. Extend the map with Substitution Method link. Justify how pattern helps simplify analysis. Interpret how this map improves solving efficiency.

2) Consider the concept map: Divide and Conquer → Problem Division → Subproblem Size → Recurrence → Complexity Explain how dividing problems leads to recurrence. Extend the map with parameters a and b. Justify their role in complexity calculation. Interpret how this connects algorithm structure and analysis

① Consider the concept Map:

Recurrence Relation  $\rightarrow$  Expansion  $\rightarrow$  Pattern Recognition  $\rightarrow$  Summation  
Final Complexity Explain how recurrence patterns are identified  
Extend the map with Substitution Method link. Justify  
how pattern helps simplify analysis. Interpret how this  
map improves solving efficiency.





- ② Consider the concept map: Divide and Conquer  $\rightarrow$  problem Division  $\rightarrow$  Subproblem size  $\rightarrow$  Recurrence  $\rightarrow$  Complexity
- Explain how recurrence patterns are identified. Extend the map with Substitution Method link. Justify how pattern helps simplify analysis. Intert how this connects algorithm structure and analysis.

